

COLUMBIA OWENS, SC, USA

WMO: 723104

Lat: 33.971N

Lon: 80.996W

Elev: 65

StdP: 100.55

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-2.1	-14.9	1.0	1.5	-12.4	1.3	3.1	8.8	13.8	8.0	12.9	0.6	320	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	36.3	24.0	35.0	23.8	33.8	23.6	25.9	32.1	25.4	31.5	25.0	30.8	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.3	27.6	23.9	18.9	27.4	23.5	18.4	27.1	80.1	32.4	77.9	31.5	76.1	30.8	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	5.8	5.0	DB	-6.7	38.2	1.8	1.8	-8.0	39.5	-9.1	40.6	-10.1	41.6	-11.4	42.9	(4)
(5)			WB	-8.1	27.0	1.7	0.7	-9.3	27.5	-10.3	27.9	-11.3	28.3	-12.5	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	8.1	10.0	13.9	18.5	22.8	26.7	28.1	27.7	24.8	19.0	13.0	9.7	(6)	
	DBStd	8.15	5.46	4.83	4.81	4.07	3.27	2.30	1.97	2.07	2.91	4.14	4.47	4.99	(7)	
	HDD10.0	268	106	55	19	1	0	0	0	0	1	19	68		(8)	
	HDD18.3	1238	320	235	150	48	6	0	0	0	1	43	167	269	(9)	
	CDD10.0	3397	45	56	141	256	398	501	560	548	444	278	110	60	(10)	
	CDD18.3	1327	2	3	14	53	145	251	302	289	195	63	8	3	(11)	
	CDH23.3	13050	10	21	144	540	1356	2527	3258	2988	1672	468	55	12	(12)	
(13)	CDH26.7	5712	1	1	24	157	533	1198	1596	1397	683	119	4	0	(13)	
(14)	Wind	WSAvg	1.7	1.9	1.9	2.1	2.1	1.8	1.7	1.6	1.3	1.4	1.3	1.3	1.6	(14)
(15) Precipitation	PrecAvg	1171	107	96	116	75	82	128	137	114	97	71	70	80	(15)	
	PrecMax	1677	240	218	254	186	188	320	379	264	260	302	189	249	(16)	
	PrecMin	778	19	16	13	6	17	22	25	21	0	0	9	11	(17)	
	PrecStd	194	54	47	58	47	41	68	82	69	62	63	43	48	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	26.1	29.0	32.2	35.3	37.4	37.9	37.7	35.1	31.8	27.3	24.9	(19)	
		MCWB	18.1	18.1	18.0	20.1	21.6	23.7	24.8	24.3	22.4	23.3	19.4	18.6	(20)	
	2%	DB	21.7	23.1	27.0	30.2	33.0	35.5	36.3	35.9	33.7	29.6	25.0	22.5	(21)	
		MCWB	16.2	15.9	17.4	19.3	21.3	23.4	24.5	24.4	22.5	21.0	17.8	17.8	(22)	
	5%	DB	19.3	21.1	24.8	28.6	31.5	33.9	34.9	34.2	32.3	27.8	22.9	20.3	(23)	
		MCWB	14.8	14.6	16.1	18.6	21.1	23.4	24.3	24.2	22.4	20.1	16.8	16.8	(24)	
	10%	DB	17.2	18.7	22.5	26.9	29.8	32.5	33.5	32.8	30.9	26.1	21.1	18.2	(25)	
		MCWB	13.3	13.7	14.9	17.8	20.6	23.0	24.2	23.9	22.2	19.5	16.4	15.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	19.5	20.1	22.8	24.3	25.8	26.8	26.6	25.3	24.6	21.7	21.0	(27)	
		MCDB	22.8	22.8	24.9	29.4	30.1	32.4	33.6	32.8	30.4	28.4	24.0	23.0	(28)	
	2%	WB	17.7	18.0	18.8	21.5	23.3	25.1	26.0	25.9	24.6	23.5	20.0	19.1	(29)	
		MCDB	20.4	21.7	24.0	27.0	28.6	31.6	32.6	31.8	29.4	27.0	22.7	21.0	(30)	
	5%	WB	16.0	16.3	17.8	20.4	22.6	24.6	25.5	25.5	24.1	22.5	18.8	17.6	(31)	
		MCDB	18.6	19.9	22.6	25.4	27.9	30.7	31.9	31.2	28.5	25.8	21.4	19.9	(32)	
	10%	WB	14.0	14.6	16.7	19.3	21.8	24.1	25.1	25.0	23.5	21.1	17.4	15.5	(33)	
		MCDB	16.3	17.8	20.8	24.1	27.2	29.7	31.1	30.3	27.7	24.5	20.4	18.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.5	12.4	12.6	11.6	10.6	10.4	9.9	10.1	11.4	12.2	10.9	(35)	
		MCDBR	13.4	14.1	14.8	14.6	13.7	12.4	12.0	11.8	12.1	12.5	13.8	12.2	(36)	
		MCWBR	8.6	8.3	6.7	5.6	4.5	3.5	3.3	3.1	3.5	5.3	7.7	8.1	(37)	
	5% WB	MCDBR	11.2	11.9	12.1	11.4	10.7	10.5	10.4	10.1	9.3	8.8	10.0	10.7	(38)	
		MCWBR	8.6	8.3	6.8	5.7	4.1	3.5	3.5	3.1	3.3	4.6	6.9	8.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.324	0.333	0.371	0.412	0.477	0.510	0.556	0.528	0.444	0.402	0.351	0.321	(40)	
	taud		2.489	2.461	2.382	2.259	2.125	2.075	1.969	2.038	2.257	2.342	2.444	2.533	(41)	
	Ebn at Noon		887	916	903	876	820	790	750	765	818	826	845	870	(42)	
	Edh at Noon		86	98	115	136	157	165	182	167	129	109	89	78	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.77	3.40	4.48	5.64	6.14	6.40	6.08	5.53	4.76	3.98	3.05	2.42	(44)	
	RadStd		0.14	0.36	0.33	0.41	0.48	0.45	0.40	0.27	0.41	0.48	0.24	0.22	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+326	+189	(46)
(47)	Regional (62 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+133	(47)

Nomenclature: See separate page